

Question 1 – Student Marks Array

The marks obtained by five students in a subject are given as [78, 65, 89, 56, 92]. Create a NumPy array and display the array along with its basic properties.

Program

```
import numpy as np
marks = np.array([78, 65, 89, 56, 92])
print("Marks:", marks)
print("Number of elements:", marks.size)
print("Shape:", marks.shape)
print("Data type:", marks.dtype)
print("Number of dimensions:", marks.ndim)
```

Output

main.py × +

1 import numpy as np

2 marks = np.array([78, 65, 89, 56, 92])

3 print("Marks:", marks)

4 print("Number of elements:", marks.size)

5 print("Shape:", marks.shape)

6 print("Data type:", marks.dtype)

7 print("Number of dimensions:", marks.ndim)

Console I/O AI Agent

Marks: [78 65 89 56 92]
Number of elements: 5
Shape: (5,)
Data type: int64
Number of dimensions: 1

Question 2 – Student Marks Access

The marks of five students are stored in a NumPy array as [72, 85, 64, 90, 76]. Write a program to access and display specific student marks using NumPy indexing and slicing.

Program

```
import numpy as np
marks = np.array([72, 85, 64, 90, 76])
print("Marks:", marks)
print("First student:", marks[0])
print("Third student:", marks[2])
print("Last student:", marks[-1])
```

```
print("First three students:", marks[:3])
print("Second to fourth students:", marks[1:4])
```

Output

main.py × +

```
1 import numpy as np
2 marks = np.array([72, 85, 64, 90, 76])
3 print("Marks:", marks)
4 print("First student:", marks[0])
5 print("Third student:", marks[2])
6 print("Last student:", marks[-1])
7 print("First three students:", marks[:3])
8 print("Second to fourth students:", marks[1:4])
```

Console I/O AI Agent 344 ms

Marks: [72 85 64 90 76]
First student: 72
Third student: 64
Last student: 76
First three students: [72 85 64]
Second to fourth students: [85 64 90]

Question 3 – Subject-wise Marks

The marks obtained by five students in three subjects are given below. Create a NumPy array to represent the data and reshape it into an appropriate matrix format. [78, 85, 90, 65, 72, 80, 88, 91, 84, 56, 62, 70, 95, 89, 92]

Program

```
import numpy as np
marks = np.array([
    78, 85, 90,
    65, 72, 80,
    88, 91, 84,
    56, 62, 70,
    95, 89, 92
])
matrix = marks.reshape(5, 3)
print("Original array:")
print(marks)
print("\nMarks matrix:")
print(matrix)
```

Output

main.py × +

```
1 import numpy as np
2 marks = np.array([
3     78, 85, 90,
4     65, 72, 80,
5     88, 91, 84,
6     56, 62, 70,
7     95, 89, 92
8 ])
9 matrix = marks.reshape(5, 3)
10 print("Original array:")
11 print(marks)
12 print("\nMarks matrix:")
13 print(matrix)
```

Console I/O AI Agent

Original array:
[78 85 90 65 72 80 88 91 84 56 62 70 95 89 92]

Marks matrix:
[[78 85 90]
 [65 72 80]
 [88 91 84]
 [56 62 70]
 [95 89 92]]

Question 4 – Internal and External Marks

The internal and external examination marks of five students are stored in two NumPy arrays. Write a program to calculate the final marks of each student using NumPy array operations.

Program

```
import numpy as np
internal = np.array([20, 18, 22, 19, 21])
external = np.array([65, 70, 60, 68, 72])
final_marks = internal + external
print("Internal marks:", internal)
print("External marks:", external)
print("Final marks:", final_marks)
```

Output

main.py × +

```
1 import numpy as np
2 internal = np.array([20, 18, 22, 19, 21])
3 external = np.array([65, 70, 60, 68, 72])
4 final_marks = internal + external
5 print("Internal marks:", internal)
6 print("External marks:", external)
7 print("Final marks:", final_marks)
```

Console I/O AI Agent

Internal marks: [20 18 22 19 21]
External marks: [65 70 60 68 72]
Final marks: [85 88 82 87 93]

Question 5 – Pass Percentage Analysis

The marks obtained by five students are [45, 78, 56, 32, 91]. Using NumPy Boolean masking, identify the students who have secured 50 marks or above.

Program

```
import numpy as np
marks = np.array([45, 78, 56, 32, 91])
passed = marks[marks >= 50]
print("Marks:", marks)
print("Students scoring 50 or above:", passed)
```

Output

<pre>main.py × + 1 import numpy as np 2 marks = np.array([45, 78, 56, 32, 91]) 3 passed = marks[marks >= 50] 4 print("Marks:", marks) 5 print("Students scoring 50 or above:", passed)</pre>	Console I/O AI Agent <pre>Marks: [45 78 56 32 91] Students scoring 50 or above: [78 56 91]</pre>
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Question 6 – Average Marks

The marks of five students in three subjects are represented using a NumPy matrix. Write a program to calculate the average marks of each student.

Program

```
import numpy as np
marks = np.array([
    [78, 85, 90],
    [65, 72, 80],
    [88, 91, 84],
    [56, 62, 70],
    [95, 89, 92]
])
average = np.mean(marks, axis=1)
print("Marks matrix:")
print(marks)
print("\nAverage marks of each student:")
print(average)
```

Output

main.py × +

```
1 import numpy as np
2 marks = np.array([
3     [78, 85, 90],
4     [65, 72, 80],
5     [88, 91, 84],
6     [56, 62, 70],
7     [95, 89, 92]
8 ])
9 average = np.mean(marks, axis=1)
10 print("Marks matrix:")
11 print(marks)
12 print("\nAverage marks of each student:")
13 print(average)
```

Console I/O AI Agent 245 ms

Marks matrix:
[[78 85 90]
[65 72 80]
[88 91 84]
[56 62 70]
[95 89 92]]

Average marks of each student:
[84.33333333 72.33333333 87.66666667 62.66666667 92.]

Question 7 – Class Performance Statistics

The marks obtained by five students are [67, 82, 91, 74, 58]. Using NumPy statistical functions, determine the total, average, highest, lowest, and standard deviation of the marks.

Program

```
import numpy as np
marks = np.array([67, 82, 91, 74, 58])
print("Marks:", marks)
print("Total:", np.sum(marks))
print("Average:", np.mean(marks))
print("Highest:", np.max(marks))
print("Lowest:", np.min(marks))
print("Standard deviation:", np.std(marks))
```

Output

main.py × +

```
1 import numpy as np
2 marks = np.array([67, 82, 91, 74, 58])
3 print("Marks:", marks)
4 print("Total:", np.sum(marks))
5 print("Average:", np.mean(marks))
6 print("Highest:", np.max(marks))
7 print("Lowest:", np.min(marks))
8 print("Standard deviation:", np.std(marks))
```

Console I/O AI Agent

Marks: [67 82 91 74 58]
Total: 372
Average: 74.4
Highest: 91
Lowest: 58
Standard deviation: 11.46472851837321

Question 8 – Subject-wise Performance

The marks of five students in three subjects are stored in a NumPy matrix. Write a program to calculate the total marks obtained in each subject using an appropriate axis operation.

Program

```

import numpy as np
marks = np.array([
    [78, 85, 90],
    [65, 72, 80],
    [88, 91, 84],
    [56, 62, 70],
    [95, 89, 92]
])
subject_total = np.sum(marks, axis=0)
print("Marks matrix:")
print(marks)
print("\nTotal marks in each subject:")
print(subject_total)

```

Output

Upgrade

AI

Python

Run

main.py

+

```

1  import numpy as np
2  marks = np.array([
3      [78, 85, 90],
4      [65, 72, 80],
5      [88, 91, 84],
6      [56, 62, 70],
7      [95, 89, 92]
8  ])
9  subject_total = np.sum(marks, axis=0)
10 print("Marks matrix:")
11 print(marks)
12 print("\nTotal marks in each subject:")
13 print(subject_total)
14

```

Console

I/O

AI Agent

Marks matrix:

```

[[78 85 90]
 [65 72 80]
 [88 91 84]
 [56 62 70]
 [95 89 92]]

```

Total marks in each subject:

```

[382 399 416]

```

Question 9 – Student-wise Performance

The marks of five students in three subjects are stored in a NumPy matrix. Write a program to calculate the total marks obtained by each student using an appropriate axis operation.

Program

```

import numpy as np
marks = np.array([
    [78, 85, 90],
    [65, 72, 80],
    [88, 91, 84],
    [56, 62, 70],
    [95, 89, 92]
])
student_total = np.sum(marks, axis=1)
print("Marks matrix:")
print(marks)

```

```
print("\nTotal marks of each student:")
print(student_total)
```

Output

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AI Python Run

main.py

```
1 import numpy as np
2 marks = np.array([
3     [78, 85, 90],
4     [65, 72, 80],
5     [88, 91, 84],
6     [56, 62, 70],
7     [95, 89, 92]
8 ])
9 student_total = np.sum(marks, axis=1)
10 print("Marks matrix:")
11 print(marks)
12 print("\nTotal marks of each student:")
13 print(student_total)
```

Console I/O AI Agent

Marks matrix:
[[78 85 90]
[65 72 80]
[88 91 84]
[56 62 70]
[95 89 92]]

Total marks of each student:
[253 217 263 188 276]

Question 10 – Student Ranking

The total marks obtained by five students are [245, 278, 219, 290, 256]. Use NumPy sorting and indexing operations to arrange the marks in order and determine the ranking of the students.

Program

```
import numpy as np
marks = np.array([245, 278, 219, 290, 256])
sorted_marks = np.sort(marks)[::-1]
ranking = np.argsort(marks)[::-1]
print("Original marks:", marks)
print("Marks in descending order:", sorted_marks)
print("\nStudent ranking:")
for rank, index in enumerate(ranking, start=1):
    print("Rank", rank, "- Student", index + 1, "-", marks[index])
```

Output

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AI

Python

Run

main.py

```
1 import numpy as np
2 marks = np.array([245, 278, 219, 290, 256])
3 sorted_marks = np.sort(marks)[::-1]
4 ranking = np.argsort(marks)[::-1]
5 print("Original marks:", marks)
6 print("Marks in descending order:", sorted_marks)
7 print("\nStudent ranking:")
8 for rank, index in enumerate(ranking, start=1):
9     print("Rank", rank, "- Student", index + 1, "-", marks[index])
```

Console

I/O

AI Agent

Original marks: [245 278 219 290 256]
Marks in descending order: [290 278 256 245 219]

Student ranking:
Rank 1 - Student 4 - 290
Rank 2 - Student 2 - 278
Rank 3 - Student 5 - 256
Rank 4 - Student 1 - 245
Rank 5 - Student 3 - 219

Question 11 – Duplicate Marks Analysis

The marks obtained by five students are [85, 92, 85, 76, 92]. Use NumPy functions to identify the unique marks obtained by the students.

Program

```
import numpy as np
marks = np.array([85, 92, 85, 76, 92])
unique_marks = np.unique(marks)
print("Marks:", marks)
print("Unique marks:", unique_marks)
```

Output

main.py

```
1 import numpy as np
2 marks = np.array([85, 92, 85, 76, 92])
3 unique_marks = np.unique(marks)
4 print("Marks:", marks)
5 print("Unique marks:", unique_marks)
```

Console

I/O

AI Agent

Marks: [85 92 85 76 92]
Unique marks: [76 85 92]

Question 12 – Missing Marks

The marks of five students are represented as [78, 85, np.nan, 92, 67], where np.nan represents a missing mark. Write a NumPy program to calculate the average marks without considering the missing value.

Program

```
import numpy as np
marks = np.array([78, 85, np.nan, 92, 67])
```

```
average = np.nanmean(marks)
print("Marks:", marks)
print("Average without missing value:", average)
```

Output

<pre>main.py × + 1 import numpy as np 2 marks = np.array([78, 85, np.nan, 92, 67]) 3 average = np.nanmean(marks) 4 print("Marks:", marks) 5 print("Average without missing value:", average)</pre>	Console I/O AI Agent <pre>Marks: [78. 85. nan 92. 67.] Average without missing value: 80.5</pre>
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Question 13 – Grade Classification

The marks obtained by five students are [95, 82, 74, 61, 45]. Using NumPy conditional operations, classify the students into appropriate grade categories based on their marks.

Program

```
import numpy as np
marks = np.array([95, 82, 74, 61, 45])
grades = np.select(
    [
        marks >= 90,
        marks >= 80,
        marks >= 70,
        marks >= 60
    ],
    [
        "A",
        "B",
        "C",
        "D"
    ],
    default="F"
)
print("Marks:", marks)
print("Grades:", grades)
```

Output

main.py × +

```
1 import numpy as np
2 marks = np.array([95, 82, 74, 61, 45])
3 grades = np.select(
4     [
5         marks >= 90,
6         marks >= 80,
7         marks >= 70,
8         marks >= 60
9     ],
10    [
11        "A",
12        "B",
13        "C",
14        "D"
15    ],
16    default="F"
17 )
18 print("Marks:", marks)
19 print("Grades:", grades)
```

Console I/O AI Agent

Marks: [95 82 74 61 45]
Grades: ['A' 'B' 'C' 'D' 'F']

Question 14 – Random Marks Generation

Generate marks for five students using NumPy's random number generation functionality. Perform basic statistical analysis on the generated marks.

Program

```
import numpy as np
np.random.seed(10)
marks = np.random.randint(40, 101, size=5)
print("Randomly generated marks:", marks)
print("Total:", np.sum(marks))
print("Average:", np.mean(marks))
print("Highest:", np.max(marks))
print("Lowest:", np.min(marks))
print("Standard deviation:", np.std(marks))
```

Output

main.py × +

```
1 import numpy as np
2 np.random.seed(10)
3 marks = np.random.randint(40, 101, size=5)
4 print("Randomly generated marks:", marks)
5 print("Total:", np.sum(marks))
6 print("Average:", np.mean(marks))
7 print("Highest:", np.max(marks))
8 print("Lowest:", np.min(marks))
9 print("Standard deviation:", np.std(marks))
```

Console I/O AI Agent

Randomly generated marks: [49 76 55 40 89]
Total: 309
Average: 61.8
Highest: 89
Lowest: 40
Standard deviation: 18.037738217415175

Question 15 – Student Performance Analysis

The marks of five students in three subjects are stored in a NumPy array. Develop a program to perform a complete student performance analysis by calculating the total marks, average marks, highest marks, lowest marks, and identifying students who perform above the class average.

Program

```
import numpy as np
marks = np.array([
    [78, 85, 90],
    [65, 72, 80],
    [88, 91, 84],
    [56, 62, 70],
    [95, 89, 92]
])
total = np.sum(marks, axis=1)
average = np.mean(marks, axis=1)
highest = np.max(marks, axis=1)
lowest = np.min(marks, axis=1)
class_average = np.mean(total)
above_average = np.where(total > class_average)[0] + 1
print("Marks matrix:")
print(marks)
print("\nTotal marks:", total)
print("Average marks:", average)
print("Highest marks:", highest)
print("Lowest marks:", lowest)
print("\nClass average total:", class_average)
print("Students above class average:", above_average)
```

Output

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main.py

```
1 import numpy as np
2 marks = np.array([
3     [78, 85, 90],
4     [65, 72, 80],
5     [88, 91, 84],
6     [56, 62, 70],
7     [95, 89, 92]
8 ])
9 total = np.sum(marks, axis=1)
10 average = np.mean(marks, axis=1)
11 highest = np.max(marks, axis=1)
12 lowest = np.min(marks, axis=1)
13 class_average = np.mean(total)
14 above_average = np.where(total > class_average)[0] + 1
15 print("Marks matrix:")
16 print(marks)
17 print("\nTotal marks:", total)
18 print("Average marks:", average)
19 print("Highest marks:", highest)
20 print("Lowest marks:", lowest)
21 print("\nClass average total:", class_average)
22 print("Students above class average:", above_average)
```

Console I/O AI Agent 29

Marks matrix:
[[78 85 90]
 [65 72 80]
 [88 91 84]
 [56 62 70]
 [95 89 92]]

Total marks: [253 217 263 188 276]
Average marks: [84.33333333 72.33333333 87.66666667 62.66666667 92.
]
Highest marks: [90 80 91 70 95]
Lowest marks: [78 65 84 56 89]

Class average total: 239.4
Students above class average: [1 3 5]